Cointegration-Based Strategies in Forex Pairs Trading

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Abstract

Pairs trading exploits pricing differentials between related assets. A cointegration-based pairs strategy selects pairs that share a statistically significant long-run relationship, so that short-term divergences are more plausibly mean-reverting. This paper studies whether that filter improves risk-adjusted performance in the foreign-exchange market. Using daily spot prices for the seven most liquid USD crosses over 2007–2025, we compare a simple always-trade pairs rule with an Engle–Granger cointegration screen on rolling 257/21-day train/test windows, across z-score entry thresholds ± 1, ± 2, and ± 3. Charging a 2 bp round-trip cost on position changes, the unlevered cointegration portfolio still delivers higher Sharpe, Sortino, and Calmar ratios than the simple benchmark at every threshold, with the largest Sharpe advantage at ± 1. The ranking is unchanged relative to frictionless (zero-cost) results and survives a grid up to 5 bp.

Keywords: Cointegration, Pairs Trading, Mean Reversion, Forex, Currency Markets

1 Introduction

Traders continually search for relative-value inefficiencies. Pairs trading is a leading example: identify assets that usually move together, and trade temporary divergences in anticipation of realignment (Gatev et al., 2006; Krauss, 2017).

Conventional implementations lean on correlation or distance statistics and treat mean reversion as a short-horizon regularity. Cointegration reframes the problem. If two non-stationary price series share a stationary linear combination, deviations from that equilibrium are statistically constrained and therefore more credible as trading signals (Engle & Granger, 1987; Vidyamurthy, 2004).

Despite a large equity literature, cointegration-based pairs trading remains comparatively thin in FX. This paper fills that gap for major currencies. The seven most liquid USD crosses are natural candidates for relative-value analysis: they share a common dollar numeraire, are tightly linked by trade and capital flows, and are disciplined by parity relations such as purchasing-power parity and interest-rate parity (Engel & West, 2005; Fama, 1984; Rogoff, 1996). Those links make long-run co-movement among majors **plausible**; they do not, by themselves, deliver a trading rule. We therefore keep the empirical design statistical: we compare (i) a **simple** pairs strategy that always trades the synthetic spread on every out-of-sample block with (ii) a **cointegration-based** strategy that trades only when Engle–Granger cointegration is detected on the preceding training window. The universe comprises all unordered pairs among seven USD-denominated majors ($C(7, 2) = 21$ pairs). Performance is evaluated with annualized return and volatility, Sharpe, Sortino, and Calmar ratios, and maximum drawdown (Sharpe, 1966; Sortino & Meer, 1991; Young, 1991) on **unlevered** portfolios. Cumulative-return figures scale the cointegration path to equal ex-post volatility for visual comparison only.

Portfolio metrics and Figure 3–Figure 5 below are produced by the companion replication code on a frozen Yahoo Finance sample.

2 Literature review

(Krauss, 2017) organizes pairs-trading research into distance, time-series, stochastic-control, cointegration, and other approaches.

The distance method of (Gatev et al., 2006) forms pairs by historical Euclidean proximity and opens when spreads widen. It is simple and relatively robust to data-snooping, but the static L^2 metric is outlier-sensitive and poorly adapted to regime shifts.

(Elliott et al., 2005) model the spread as a mean-reverting Gaussian Markov chain in state space. The framework is tractable, yet Gaussian assumptions can fail in FX, especially in stress episodes.

(Jurek & Yang, 2007) study stochastic control for arbitrageurs choosing between a mean-reverting spread and a risk-free asset, obtaining closed-form policies under Ornstein–Uhlenbeck uncertainty.

(Vidyamurthy, 2004) provides the cointegration blueprint: pair preselection, Engle–Granger tradability screening, and nonparametric entry/exit rules. (Rad et al., 2015) implement distance preselection plus Engle–Granger and find return profiles similar to pure distance methods. (Galenko et al., 2012) use multivariate cointegration weights on ETF baskets and document mean-reverting portfolio returns, with the usual caveat that performance can weaken outside the estimation regime.

Related applications include equity hedging via cointegration (Burgess, 2003), gold–silver parity (Liu & Chou, 2003), inflation hedges (Bampinas & Panagiotidis, 2015; 2016), and currency-portfolio optimization (Dunis et al., 2011). (Huck & Afawubo, 2015) compare distance, stationarity, and cointegration on S&P 500 constituents and conclude that cointegration delivers more stable excess returns after costs; our contribution is the analogous horse race in major FX. Ranking/forecasting approaches (Huck, 2009; 2010) are complementary but outside our design.

Relative to short-horizon correlation filters, cointegration anchors trades in a long-run equilibrium. In FX that equilibrium has a natural economic reading via parity and common macro drivers; we spell out that motivation—and what we do **not** estimate—in the next section.

3 Methodology

3.1 Data

We use daily adjusted closes for EUR/USD, GBP/USD, USD/JPY, USD/CHF, USD/CAD, AUD/USD, and NZD/USD from Yahoo Finance (Yahoo Finance, 2025), January 1, 2007 to December 31, 2025. Quotes with USD as numerator are inverted so every series is XXXUSD

(e.g.
$\mathrm{JPYUSD} = \frac{1}{\mathrm{USDJPY}}$) (Pound Sterling Live, 2022). Analysis uses log prices.

Prices are processed in rolling blocks: a training window of $n = 257$ days (about one year) followed by a testing window of $m = 21$ days (about one month). After each test block, windows advance by m days so the next train ends where the previous test ended (Figure 1). The training sample is used to (i) screen for cointegration when required and (ii) estimate the spread mean and standard deviation used in the out-of-sample z-score. The simple strategy skips (i) but still uses the training moments.

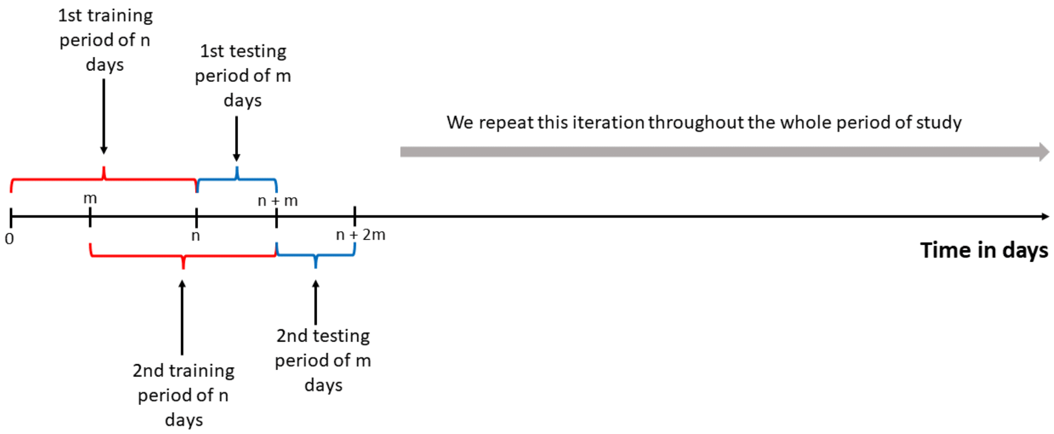

Figure 1: Iterative training and testing windows over the sample.

3.2 Economic motivation

Why should an Engle–Granger screen be more than a statistical filter among 21 correlated USD crosses? Purchasing-power parity (PPP) links national price levels to exchange rates; although real exchange rates mean-revert only slowly, the associated “PPP puzzle” still implies that related nominal rates are not free to wander independently forever (Rogoff, 1996). Covered and uncovered interest-rate parity connect spot rates, forwards, and interest differentials; even when uncovered interest parity fails as a short-horizon forecast ((Fama, 1984)), exchange rates remain asset prices tightly tied to macro fundamentals ((Engel & West, 2005)). Together with a shared USD numeraire and deep G10 liquidity, these parity and macro links make **long-run co-movement**—and therefore cointegration-based relative-value trading—economically plausible in this universe ((Dunis et al., 2011)).

Importantly, we do **not** estimate PPP baskets, trade real-exchange-rate gaps, or form positions from UIP residuals. The trading object remains a bivariate log-price spread among unordered XXXUSD majors, gated by the Engle–Granger screen described below. Parity relations motivate **where** we look; the pre-committed EG rule determines **when** we trade. Parity-based screens are left as a natural extension.

3.3 Cointegration analysis

We test long-run co-movement with the Engle–Granger procedure (Engle & Granger, 1987). Johansen methods can recover multiple relations in larger systems; for bivariate FX pairs we prefer Engle–Granger for transparency and leave Johansen to future work.

Step 1. Augmented Dickey–Fuller (ADF) tests on each log-price series. Failure to reject the unit-root null supports treating both series as $I(1)$.

Step 2. OLS long-run regression

$$y_t = \alpha + \beta x_t + \varepsilon_t \tag{1}$$

with residual

$$\hat{\varepsilon}_t = y_t - \hat{\beta} x_t - \hat{\alpha} \tag{2}$$

.

Step 3. ADF test on $\hat{\varepsilon}_t$. Rejecting the residual unit-root null is evidence of cointegration: y_t and x_t share a stationary combination even though each is non-stationary. The hedge ratio $\hat{\beta}$ defines the synthetic spread.

Cointegration is a **relation**, not a directed trading object, so we work with $C(7,2) = 21$ unordered pairs. Legs are ordered alphabetically by ticker so that each pair has a unique synthetic spread $\log P^1 - \beta \log P^2$. The Engle–Granger residual can depend on which series is the regressand; as a fixed design choice we therefore test **both** OLS orientations on each training window and, among those that pass at 5%, retain the orientation with the clearer residual ADF statistic (more negative t-stat), mapping that hedge into the alphabetical spread. If neither orientation passes, the pair is flat for the following test window.

3.4 Mean-reversion strategy

On each testing day the z-score of the spread is

$$Z_t = \frac{\hat{\varepsilon}_t - \mu}{\sigma} \tag{3}$$

where μ and σ are the training mean and standard deviation of $\hat{\varepsilon}_t = \log P_t^1 - \beta \log P_t^2$ (intercept cancels in the z-score).

Enter a long spread when $Z_t < -z^\star$ and a short spread when $Z_t > z^\star$, for thresholds $z^\star \in \{1, 2, 3\}$. Signals are lagged two days to allow for execution delay:

$$R_t^{\text{gross}} = \text{Signal}_{t-2} \times (r_t^1 - r_t^2) \tag{4}$$

with equal notional on each leg (not β-hedged PnL). Flat days enter the Sharpe denominator as zeros.

Transaction costs. Let $s_t \in \{-1, 0, 1\}$ denote the lagged signal. We charge a round-trip cost of κ basis points of pair notional on position changes,

$$\text{cost}_t = (\kappa/10^4) \times |s_t - s_{t-1}| \ /2 \tag{5}$$

, so that opening or closing ($|\Delta s|$=1) costs $\kappa/2$ bp and a long–short flip ($|\Delta s|$=2) costs a full κ bp. Net returns are

$$R_t^{\text{net}} = R_t^{\text{gross}} - \text{cost}_t \tag{6}$$

. We report the grid $\kappa \in \{0, 1, 2, 5\}$, with **headline** results at $\kappa = 2$ bp (a conventional all-in estimate for liquid G10 spot). Zero cost recovers the frictionless horse race; 5 bp is a conservative upper bound. Swap/rollover is left as a caveat (short test windows and frequent flats keep overnight exposure limited).

3.5 Strategy implementation and portfolio construction

Simple pairs. Trade every testing window for every unordered pair, using training moments only for the z-score (alphabetical OLS hedge, no cointegration gate).

Cointegration-based pairs. Trade a testing window only if the Engle–Granger screen above accepts cointegration on the preceding training window; otherwise the pair is flat.

Pair returns are summed and divided by 21 to form a standardized portfolio. Main tables report these series **unlevered** and, unless noted, **after** $\kappa = 2$ bp costs. Because the cointegration book is often flat, its unlevered volatility is much smaller than the always-trade benchmark, so raw return and drawdown levels are hard to compare by eye. As a **companion** presentation only, we also rescale each strategy ex post by $L = \sigma^\star/\hat{\sigma}$ to a common annualized volatility target $\sigma^\star = 10\%$ (Table 8). We choose 10% as a round, conventional risk budget (neither fitted to the sample nor tied to either strategy's realized vol). Sharpe and Sortino are invariant to this scale; return, maximum drawdown, and Calmar are not, because compounded drawdowns are nonlinear in L. Cumulative-return figures separately rescale the cointegration path so that **daily** volatilities match the simple path (Table 1); those figures are visual aids only.

z-score	Equal-vol scale L	Ann. vol EG (%)	Ann. vol simple (%)
± 1	4.50	0.89	4.02
± 2	4.75	0.60	2.85
± 3	5.39	0.38	2.06

Table 1: Equal ex-post daily-vol scales used in cumulative-return figures only ($\kappa = 2$ bp; replication sample).

4 Results

4.1 Descriptive analysis

Per-currency return paths and additional descriptive heatmaps are omitted for brevity. Figure 2 shows pair-level annualized Sharpe ratios at the baseline threshold $z^\star = 1$ after $\kappa = 2$ bp costs. Dispersion across unordered pairs is large; a few crosses (notably some GBP and JPY combinations in this sample) dominate the right tail of risk-adjusted pair performance.

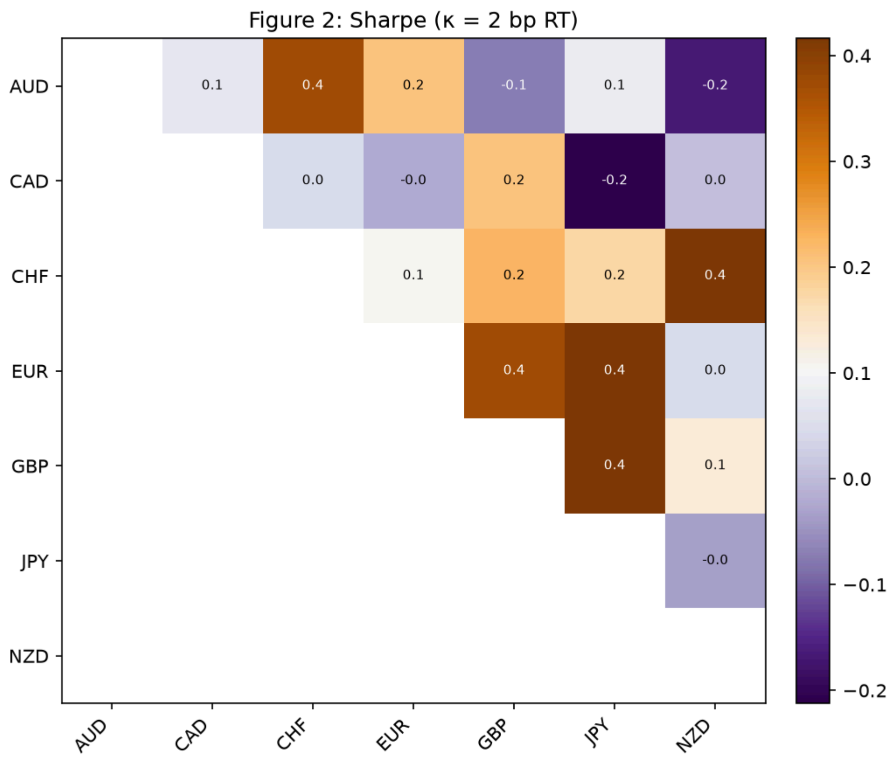

Figure 2: Heatmap of annualized Sharpe ratios by unordered pair ($z^\star = 1$, EG-filtered, $\kappa = 2$ bp RT).

4.2 Cointegration-based strategy versus simple pairs

We compare **unlevered** portfolios at training/testing windows 257/21. Headline tables use $\kappa = 2$ bp; Table 7 reports Sharpe across the full cost grid. Table 2 and Figure 3 give the $z^\star = 1$ results (figures use equal ex-post vol scaling on net returns).

Metric	Cointegration	Simple
Annualized return (%)	0.49	−0.17
Annualized volatility (%)	0.89	4.02
Sharpe ratio	0.55	−0.04
Sortino ratio	0.63	−0.05
Calmar ratio	0.20	−0.01
Maximum drawdown (%)	−2.48	−15.89

Table 2: Unlevered performance at $z^\star = \pm 1$ after $\kappa = 2$ bp RT costs (windows 257/21, 21 unordered pairs, sample through 2025). Frictionless Sharpes are 0.62 (EG) and 0.01 (simple).

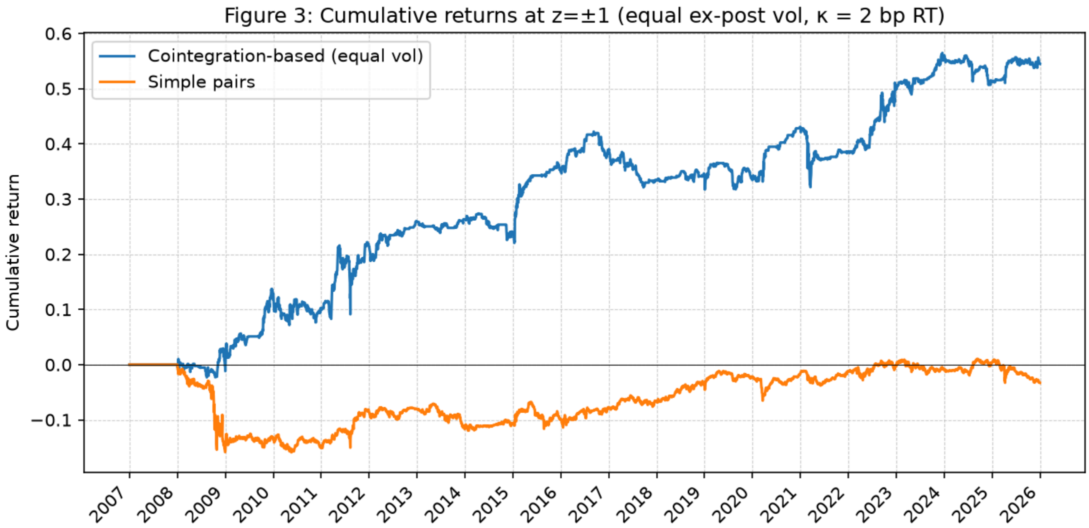

Figure 3: Cumulative returns at $z^\star = \pm 1$ (equal ex-post vol; $\kappa = 2$ bp; visuals only).

Unlevered and after costs, cointegration still earns a positive annualized return at far lower volatility and drawdown than the always-trade rule, whose 2 bp haircut turns its already thin $z^\star = 1$ edge slightly negative. The equal-vol equity curve rises more steadily than the simple path, which suffers deeper early-sample drawdowns (notably around 2008 and 2011–12).

Widening the entry band to ± 2 (Table 3, Figure 4) reduces trading intensity and absolute returns for the cointegration book, but the filter retains a clear risk-adjusted edge after costs.

Metric	Cointegration	Simple
Annualized return (%)	0.25	0.18
Annualized volatility (%)	0.60	2.85
Sharpe ratio	0.41	0.06
Sortino ratio	0.34	0.06
Calmar ratio	0.13	0.01
Maximum drawdown (%)	−1.93	−12.51

Table 3: Unlevered performance at $z^\star = \pm 2$ after $\kappa = 2$ bp RT costs.

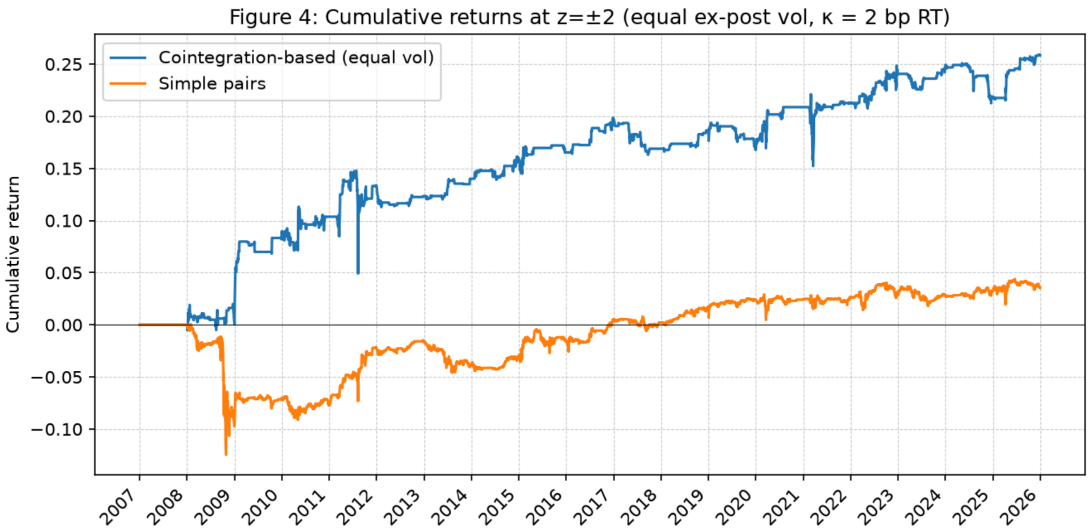

Figure 4: Cumulative returns at $z^\star = \pm 2$ (equal ex-post vol; $\kappa = 2$ bp; visuals only).

At ± 3 (Table 4, Figure 5) both books trade less often; cointegration still leads on Sharpe and related ratios after costs, with much smaller unlevered drawdowns.

Metric	Cointegration	Simple
Annualized return (%)	0.18	0.21
Annualized volatility (%)	0.38	2.06
Sharpe ratio	0.48	0.10
Sortino ratio	0.25	0.07
Calmar ratio	0.21	0.02
Maximum drawdown (%)	−0.86	−9.27

Table 4: Unlevered performance at $z^\star = \pm 3$ after $\kappa = 2$ bp RT costs.

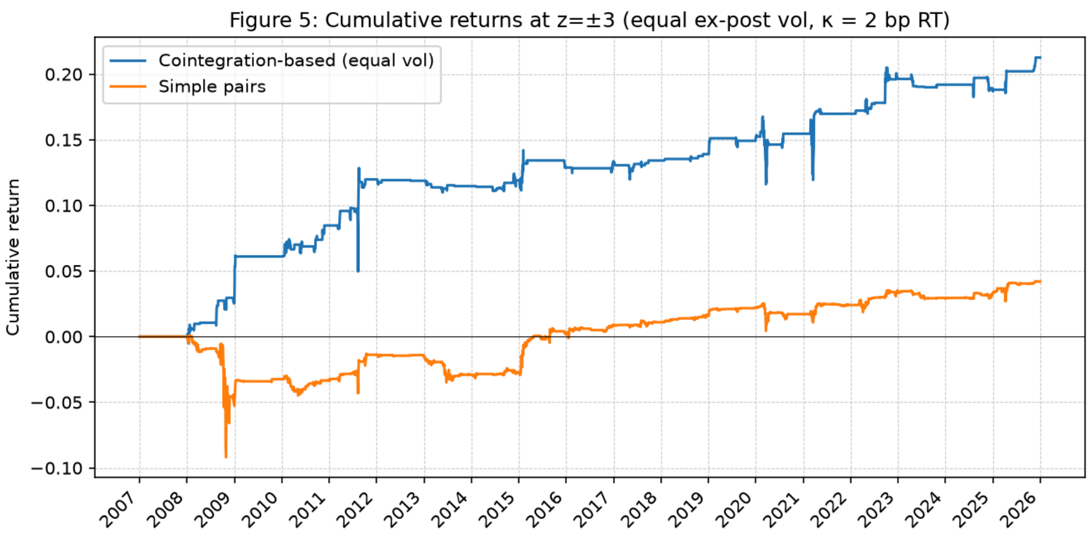

Figure 5: Cumulative returns at $z^\star = \pm 3$ (equal ex-post vol; $\kappa = 2$ bp; visuals only).

Table 5 and Table 6 summarize the threshold comparative at $\kappa = 2$. Table 7 shows that raising κ compresses both Sharpes, but the EG advantage remains positive through 5 bp at every threshold; at ± 1 the simple book is already near zero without costs and negative once $\kappa \geq 1$.

z-score	Cointegration Sharpe	Simple Sharpe
± 1	0.55	-0.04
± 2	0.41	0.06
± 3	0.48	0.10

Table 5: Unlevered Sharpe ratios across thresholds after $\kappa = 2$ bp RT costs.

z-score	Cointegration MDD (%)	Simple MDD (%)
± 1	-2.48	-15.89
± 2	-1.93	-12.51
± 3	-0.86	-9.27

Table 6: Unlevered maximum drawdowns by threshold after $\kappa = 2$ bp RT costs.

z-score	κ (bp)	EG Sharpe	Simple Sharpe
± 1	0	0.62	0.01
	1	0.58	-0.02
	2	0.55	-0.04
	5	0.45	-0.12
± 2	0	0.45	0.11
	1	0.43	0.08
	2	0.41	0.06
	5	0.35	-0.00
± 3	0	0.50	0.13
	1	0.49	0.11
	2	0.48	0.10
	5	0.43	0.06

Table 7: Unlevered Sharpe sensitivity to round-trip cost κ (bp of pair notional).

Table 8 reports the same portfolios after ex-post scaling to 10% annualized volatility ($\kappa = 2$). At matched vol, cointegration's higher Sharpe translates into higher scaled annualized return (e.g. 5.52% vs -0.43% at ± 1). Scaled maximum drawdowns are large for both books — especially cointegration, which requires substantial leverage to reach 10% vol from a low unlevered base — so Calmar need not preserve the unlevered ranking. We therefore treat Table 8 as a magnitude aid, not a replacement for the unlevered metrics.

z	Strategy	Scale L	Ann. ret. (%)	Sharpe	Calmar	MDD (%)
±1	Cointegration	11.21	5.52	0.55	0.13	−43.8
	Simple	2.49	−0.43	−0.04	−0.01	−35.0
±2	Cointegration	16.69	4.11	0.41	0.09	−43.9
	Simple	3.51	0.62	0.06	0.02	−37.5
±3	Cointegration	26.11	4.76	0.48	0.11	−41.6
	Simple	4.84	1.01	0.10	0.03	−37.7

Table 8: Companion metrics at 10% target annualized volatility after $\kappa = 2$ bp (ex-post $L = 0.10/\hat{\sigma}$). Ann. volatility is 10% by construction; Sharpe matches the unlevered net table.

5 Conclusions and future research

This study asks whether an Engle–Granger cointegration filter improves FX pairs trading among seven liquid USD crosses. On rolling 257/21 windows and 21 unordered pairs, the **unlevered** cointegration portfolio outperforms the always-trade benchmark on Sharpe, Sortino, and Calmar at $z^\star \in \{1, 2, 3\}$ after a 2 bp round-trip cost, with the strongest Sharpe edge at ±1. The ranking matches the frictionless case and remains positive through 5 bp. A companion 10% target-vol table makes return magnitudes easier to compare while leaving Sharpe unchanged. Wider thresholds further reduce unlevered volatility and drawdowns but shrink the incremental Sharpe benefit of the filter.

Natural extensions include a broader grid of window lengths and thresholds; parity-based screens (PPP or UIP residuals) as alternatives or robustness checks to Engle–Granger; accounting for cross-pair dependence when interpreting the 21-pair panel; overnight swap costs; and Johansen screens as further robustness.

Conflict of interest. The authors declare no potential conflict of interest.

References

[1] E. Gatev, W. N. Goetzmann, and K. G. Rouwenhorst, "Pairs trading: Performance of a relative-value arbitrage rule," *Review of Financial Studies*, vol. 19, no. 3, pp. 797–827, 2006, doi: 10.1093/rfs/hhj020.

[2] C. Krauss, "Statistical arbitrage pairs trading strategies: Review and outlook," *Journal of Economic Surveys*, vol. 31, no. 2, pp. 513–545, 2017, doi: 10.1111/joes.12153.

[3] R. F. Engle and C. W. J. Granger, "Co-integration and error correction: Representation, estimation, and testing," *Econometrica*, vol. 55, no. 2, pp. 251–276, 1987, doi: 10.2307/1913236.

[4] G. Vidyamurthy, *Pairs Trading: Quantitative Methods and Analysis*. John Wiley & Sons, 2004.

[7] K. Rogoff, "The purchasing power parity puzzle," *Journal of Economic Literature*, vol. 34, no. 2, pp. 647–668, 1996.

[6] E. F. Fama, "Forward and spot exchange rates," *Journal of Monetary Economics*, vol. 14, no. 3, pp. 319–338, 1984, doi: 10.1016/0304-3932(84)90046-1.

[5] C. Engel and K. D. West, "Exchange rates and fundamentals," *Journal of Political Economy*, vol. 113, no. 3, pp. 485–517, 2005, doi: 10.1086/429137.

[8] W. F. Sharpe, "Mutual fund performance," *Journal of Business*, vol. 39, no. 1, pp. 119–138, 1966, doi: 10.1086/294846.

[9] F. A. Sortino and R. van der Meer, "Downside risk," *Journal of Portfolio Management*, vol. 17, no. 4, pp. 27–31, 1991, doi: 10.3905/jpm.1991.409343.

[10] T. W. Young, "Calmar ratio: A smoother tool," *Futures*, vol. 20, no. 1, pp. 40–41, 1991.

[11] R. J. Elliott, J. Van Der Hoek, and W. P. Malcolm, "Pairs trading," *Quantitative Finance*, vol. 5, no. 3, pp. 271–276, 2005, doi: 10.1080/14697680500149370.

[12] J. W. Jurek and H. Yang, "Dynamic portfolio selection in arbitrage," 2007. [Online]. Available: https://ssrn.com/abstract=882536

[13] H. Rad, R. K. Y. Low, and R. W. Faff, "The profitability of pairs trading strategies: Distance, cointegration, and copula methods," 2015. [Online]. Available: https://ssrn.com/abstract=2614233

[14] A. Galenko, E. Popova, and I. Popova, "Trading in the presence of cointegration," *The Journal of Alternative Investments*, vol. 15, no. 1, pp. 85–97, 2012, doi: 10.3905/jai.2012.15.1.085.

[15] A. N. Burgess, "Using cointegration to hedge and trade international equities," *Applied Quantitative Methods for Trading and Investment*. Wiley, 2003.

[16] S. Liu and C.-H. Chou, "Parities and spread trading in gold and silver markets: A fractional cointegration analysis," *Applied Financial Economics*, vol. 13, no. 12, pp. 899–911, 2003, doi: 10.1080/0960310032000129626.

[17] G. Bampinas and T. Panagiotidis, "Are gold and silver a hedge against inflation? A two-century perspective," *International Review of Financial Analysis*, vol. 41, pp. 267–276, 2015, doi: 10.1016/j.irfa.2015.02.007.

[18] G. Bampinas and T. Panagiotidis, "Hedging inflation with individual US stocks: A long-run portfolio analysis," *The North American Journal of Economics and Finance*, vol. 37, pp. 374–392, 2016, doi: 10.1016/j.najef.2016.05.007.

[19] C. L. Dunis, J. Laws, and A. Shone, "Cointegration-based optimisation of currency portfolios," *Journal of Derivatives & Hedge Funds*, vol. 17, pp. 86–114, 2011, doi: 10.1057/jdhf.2011.11.

[20] N. Huck and K. Afawubo, "Pairs trading and selection methods: Is cointegration superior?," *Applied Economics*, vol. 47, no. 6, pp. 599–613, 2015, doi: 10.1080/00036846.2014.975417.

[21] N. Huck, "Pairs selection and outranking: An application to the S&P 100 index," *European Journal of Operational Research*, vol. 196, no. 2, pp. 819–825, 2009, doi: 10.1016/j.ejor.2008.03.025.

[22] N. Huck, "Pairs trading and outranking: The multi-step-ahead forecasting case," *European Journal of Operational Research*, vol. 207, no. 3, pp. 1702–1716, 2010, doi: 10.1016/j.ejor.2010.06.043.

[23] Yahoo Finance, "Historical FX spot prices." [Online]. Available: https://finance.yahoo.com/

[24] Pound Sterling Live, "USD to JPY exchange rate history." [Online]. Available: https://www.poundsterlinglive.com/history/USD-JPY-2022